Ashutosh Singh

Aspiring AI/ML Engineer

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EDUCATION

National Institute of Technology, Jamshedpur *2023 – 2027*
B.Tech in Industrial Engineering, CGPA: 7.82 *Jharkhand, India*

Lucknow Public School *2020 – 2022*
Intermediate – 86.4% | *Matriculation – 88.6%* *Lucknow, India*

SKILLS

- **Core:** Machine Learning, Deep Learning, NLP, Generative AI, Computer Vision, Data Analysis
- **AI/ML:** Regression, Classification, Clustering, Time Series Forecasting, CNN, Transfer Learning, Object Detection, Image Segmentation
- **GenAI:** LLMs, RAG, LangChain, LlamaCPP, Embeddings, Semantic Search, Prompt Engineering, Pinecone
- **Tools:** Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, OpenCV, Flask, Streamlit, MySQL, Power BI, Git, AWS
- **Soft Skills:** Teamwork, Time Management, Leadership, Effective Communication

EXPERIENCE

Xylofi AI *June 2026 – July 2026*
Data Science Intern *Certificate | Live Demo*

- Built and deployed a Sales Forecasting Dashboard, an interactive multi-page Streamlit application for sales analytics, forecasting, anomaly detection, and product demand segmentation (GitHub).
- Implemented a live Forecast Explorer using SARIMA $(2, 1, 0)(0, 1, 2, 12)$ models (via statsmodels) to generate 1–3 month sales forecasts by category and region, complete with 95% confidence intervals, MAE, and RMSE.
- Developed an Anomaly Report module combining Isolation Forest and a rolling Z-score method to flag unusual weekly sales patterns, visualized through interactive trend charts with a searchable anomaly table.
- Built a Product Demand Segmentation module using K-Means clustering on sales volume, growth rate, volatility, and average order value, with PCA-based visualization of cluster membership.
- Engineered the data pipeline with pandas and NumPy so that all forecasts, anomalies, and clusters recompute live from the underlying dataset on every filter change, and deployed the application on Streamlit Community Cloud.

PROJECTS

Real Estate App *Live Demo*

- Developed a platform featuring price prediction, property recommendation, market analysis, and insights generation for real estate data.
- Applied multiple Machine Learning algorithms and achieved the best performance using Random Forest with hyper-parameter tuning.
- Built the project using Python, Scikit-learn, Pandas, NumPy, Streamlit, Plotly, Seaborn, and Matplotlib for interactive analysis, visualization, and user experience; gained hands-on experience deploying on AWS EC2 instances.

Self-Driving Car *GitHub*

- Built an end-to-end autonomous driving perception and control pipeline combining deep learning models for steering-angle regression, lane segmentation, and surrounding-object detection.
- Developed a CNN-based steering-angle regression model inspired by NVIDIA's PilotNet and fine-tuned YOLO11 and YOLO11n-seg models for real-time lane and object segmentation.
- Implemented concurrent multi-model inference using Python's `concurrent.futures`, with OpenCV-based preprocessing and live visualization of predicted steering, lane regions, and detected objects.

End-to-End Retrieval-Augmented Generation (RAG) Medical Chatbot

- Configured localized inference of Meta's open-source LLaMA 2 model on consumer-grade hardware using LlamaCPP and LangChain, removing dependence on external LLM APIs and reducing inference costs.
- Engineered a document ingestion pipeline to split medical reference books into text chunks, generate high-dimensional vector embeddings, and index them in Pinecone for real-time semantic retrieval.
- Developed a production-ready Flask interface with a modular Python package structure using `setup.py` and helper utilities to serve context-refined answers to natural-language medical queries.

POSITIONS OF RESPONSIBILITY

- **Captain,** Team Revanta – NIT Jamshedpur
- **Joint Core,** Ojass and Culfest – NIT Jamshedpur